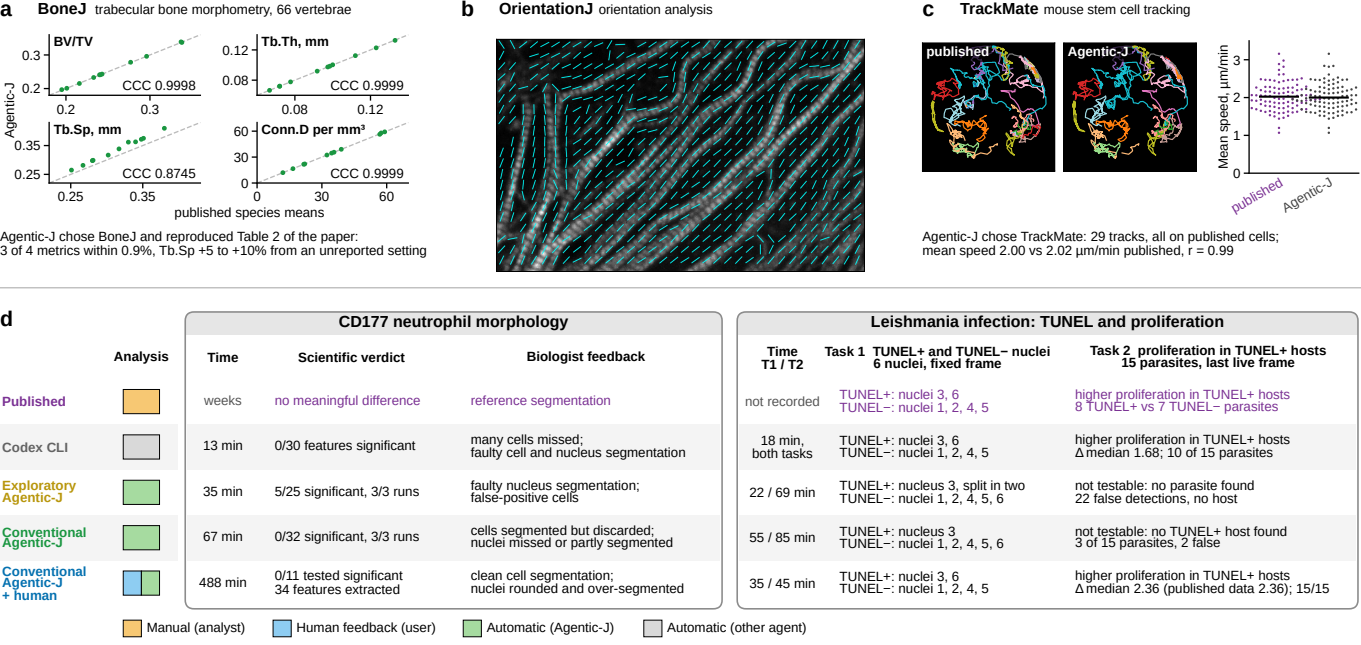


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Analysis	CD177 neutrophil morphology: Time	Scientific verdict	Biologist feedback	Leishmania infection: Time T1 / T2	Task 1 TUNEL+ and TUNEL− nuclei 6 nuclei, fixed frame	Task 2 proliferation in TUNEL+ hosts 15 parasites, last live frame
Published	weeks	no meaningful difference	reference segmentation	not recorded	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts 8 TUNEL+ vs 7 TUNEL− parasites
Codex CLI	13 min	0/30 features significant	many cells missed; faulty cell and nucleus segmentation	18 min, both tasks	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts Δ median 1.68; 10 of 15 parasites
Exploratory Agentic-J	35 min	5/25 significant, 3/3 runs	faulty nucleus segmentation; false-positive cells	22 / 69 min	TUNEL+: nucleus 3, split in two TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no parasite found 22 false detections, no host
Conventional Agentic-J	67 min	0/32 significant, 3/3 runs	cells segmented but discarded; nuclei missed or partly segmented	55 / 85 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no TUNEL+ host found 3 of 15 parasites, 2 false
Conventional Agentic-J + human	488 min	0/11 tested significant 34 features extracted	clean cell segmentation; nuclei rounded and over-segmented	35 / 45 min	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts Δ median 2.36 (published data 2.36); 15/15

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent)

